

04 — Estimation & Costing

Problem-statement deliverable: *Documentation explaining how the estimation works*

Source of truth in code: `backend/app/services/costing.py`.

1. Design principle

Costing is deterministic. Segmentation supplies pixel masks; the user supplies approximate facade dimensions and (optionally) rates. No generative model invents rupees. Every line item is explainable as quantity × rate.

2. Inputs

Input	Source	Default
Category masks (<code>pixel_area</code> , <code>instance_count</code>)	Segmentation / mask edit	—
Image size (<code>width</code> × <code>height</code>)	Ingested project image	—
Facade width & height (m)	Studio UI	12 m × 9 m
Waste factor	Engine constant	1.10 (10%)
Material <code>rate_inr</code> & <code>unit</code>	Material catalog / Cost page	Group defaults if rate ≤ 0
Material selections	Studio (category → material)	—

3. Surface / quantity model

3.1 Facade area

[
$$A_{\text{facade}} = W_{\text{facade}} \times H_{\text{facade}} \quad (\mathrm{m}^2)$$

]

Widths/heights are clamped to a minimum of **0.5 m** to avoid division-by-zero nonsense.

3.2 Area-based materials (`unit == "sqm"`)

Most facade finishes (paint, cladding, tiles, patterns):

[
$$q = \left(\frac{\text{pixel_area}}{\text{image_width} \times \text{image_height}} \right) \times A_{\text{facade}} \times w$$

]

where ($w \geq 1$) is the waste factor (default **1.10**).

- If ($0 < q < 0.25$), quantity is raised to **0.25 sqm** (avoids ■0 for tiny valid patches).
- Display unit: **sqm**.

Interpretation: The fraction of the photo covered by the mask is treated as the same fraction of the user-stated facade area. This is approximate and assumes a roughly frontal elevation.

3.3 Piece-based categories (`unit == "unit"`)

Categories: **gate, door, window, pillar**.

```
[  
q = \max(1, \text{instance\_count})  
]
```

Display unit: **pcs**.

3.4 What is *not* computed (yet)

- Paint volume in litres (no coats × coverage curve)
- Tile packing by piece size
- Railing length in running metres
- Separate labour vs material columns

Those are natural extensions of the same RegionMap (see Limitations).

4. Cost formula

For each selection line:

```
[  
\text{line\_total} = q \times \text{rate\_inr}  
]
```

```
[  
\text{grand total} = \sum \text{line\_totals}  
]
```

Subtotal equals total in the current build (no tax / contingency line beyond the waste factor).

Currency: **INR (■)**. PDF uses `Rs.` for font glyph safety.

5. Default rates (■)

Used when the stored rate is missing or ≤ 0 :

Material group	Default rate	Typical unit
paint	55	■/sqm

cladding	2200	■/sqm
tiles	1100	■/sqm
patterns	1500	■/sqm
texture (fallback)	1200	■/sqm

Users override these on `/cost` via PUT `/api/materials/rates`. Recalculation uses the new rates on the next estimate — no re-segmentation required.

6. Worked example

- Image: 1200 × 900 px → 1 080 000 px
- Facade: 12 m × 9 m → **108 m²**
- Wall mask: 270 000 px → fraction **0.25**
- Material: exterior emulsion (paint) at ■55/sqm
- Waste: 1.10

```
[
q = 0.25 \times 108 \times 1.10 = 29.7\;\mathrm{sqm}
]
```

```
[
\text{line} = 29.7 \times 55 = \text{■}1{,}633.50
]
```

Add other category lines (balcony cladding, gate pieces, ...) for the grand total.

7. Live estimate vs PDF

Path	Endpoint	Behaviour
Live studio panel	POST <code>/api/costing/{project_id}/estimate</code>	JSON line items + totals + disclaimer
PDF report	POST <code>/api/reporting/{project_id}/pdf</code>	Same logic embedded with images (requires prior render)

8. Disclaimer (shown in API and PDF)

Approximate estimate based on selected finishes, detected mask coverage, and rates from the Cost page.
 Not a formal quotation.

This matches problem-statement Section 7: cost estimates are **advisory**.